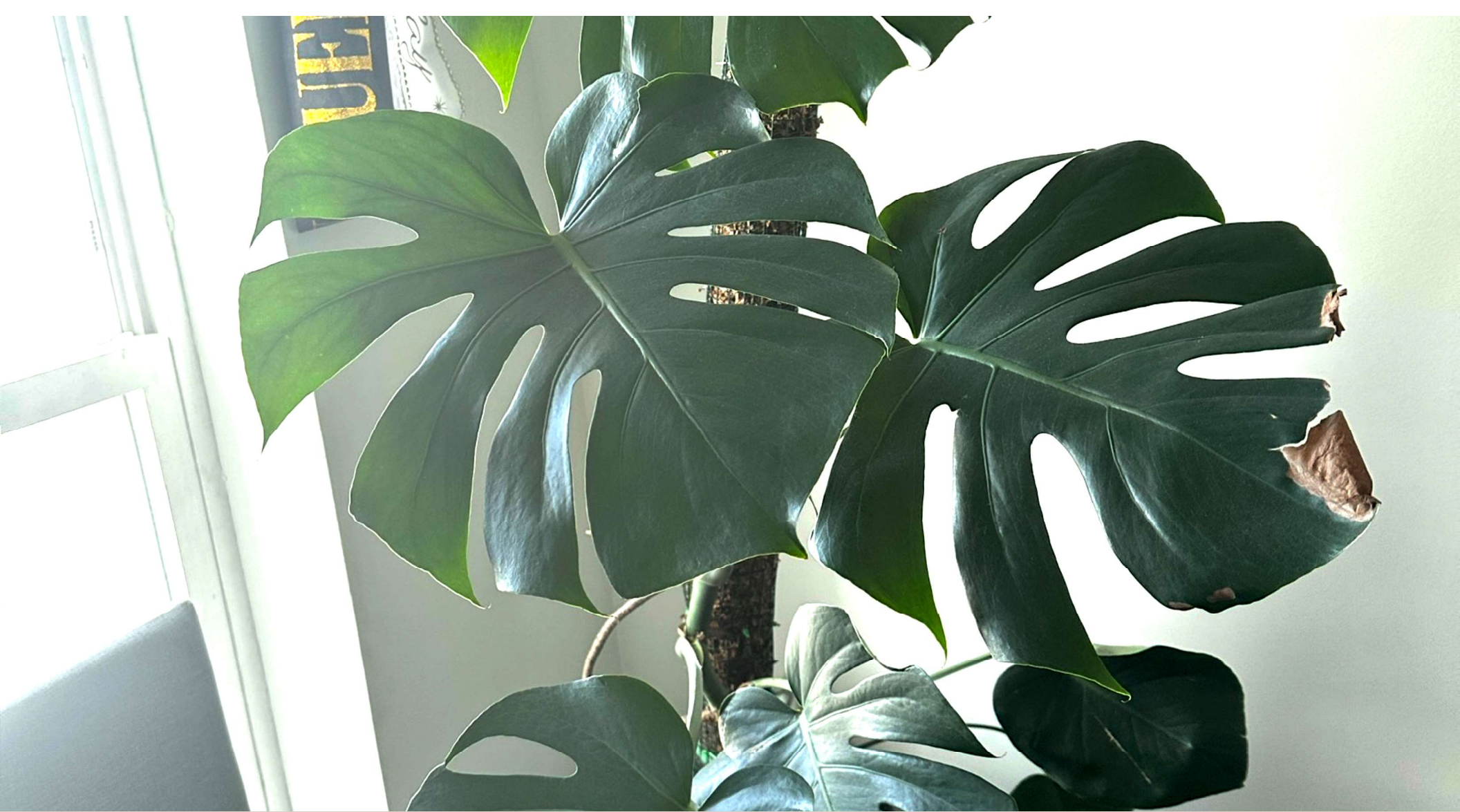




Monstera Division

And Moss Pole Introduction



Section 01

Get your shit together



01 Get your shit together

Setting up

We've got a lot to cover! But first and foremost, we need to make sure we have a workable space and our supplies are ready to go. There's nothing worse than getting 2 hours into this and realizing you forgot to mix your soil. So here's a checklist of things to make sure you have ready:

1. Set the space. You'll need a fair amount of room to spread out and be prepared to make a mess. So lay some towels or a sheet out, whatever you've got, and define your working space. Get all of your tools together and keep everything in one spot in this space so that it's all together.

2. Mix your soil now. Grab yourself a big tub (if you've got a shipping box, just use that) and go ahead and empty all of your soil, perlite, and bark into it. Give it a nice thorough mixing and then set it aside. We won't need this until around the end, but it's nice to have it done and ready because you're gonna be tired by the time we get there.

3. Build your moss poles now. Again, we won't need this until the end but you'll be very happy if you just prepare them and get it out the way now. If you have the DIY Moss Pole kit we recommend, it'll be pretty self explanatory but because this is a guide, we're going to spell out everything so get ready for some math.

You want a nice thick pole that can support a *Deliciosa* weight (and they get heavy). They may be small now, but they grow like wildfire and it's extremely difficult to repair a moss pole that was made too thin - once the plant is rooted in it, it's near impossible. So let's get it right the first time. You want it to be at least 4" thick. To figure out how much wiring to cut, we take the formula $C=\pi d$ and plug in what we have so $C=\pi(4)$. $C\sim 12"$, so you'll need to cut a 12" long piece of wiring and then use the zip ties to form it into a cylinder. There, math is over.

Please use the gloves included in the kit when working with the wiring, these things are insanely sharp. Use a zip tie every 4-5" as you attach the sides together to ensure no moss can fall out and build as many as you think you may need; *Monstera*s are often sold with several plants in a pot so count them up and build as many poles. If you're using the kit we recommend, the wiring is 8.3" tall which should be fine for the beginning of your plants and should yield 3 poles per roll. But take note, you'll repeat this process every time you need to extend it and use the same 12" cut to maintain the circumference as you build upwards - we had to extend ours 3 times in one growing season. It'll look too large for your small plants, but we need to start it right and not play catch up later.

01 Get your shit together

Tips and Tools

Grab yourself a chopstick and keep it nearby. We'll be referencing using one often, and it's an incredibly useful tool for divisions. The cheap wooden ones from your local Chinese restaurant is perfect.

To the right is my working space for a regular repot. You'll need a larger one because you'll be working with many plants, but note that all of my tools are on the mat and I try to keep everything contained to the area.

This isn't for cleanliness. Once we start dividing and laying plants down, we need to make sure they're in a safe spot. I found myself tossing them wherever I could and ended up stepping on a few - try not to do that.



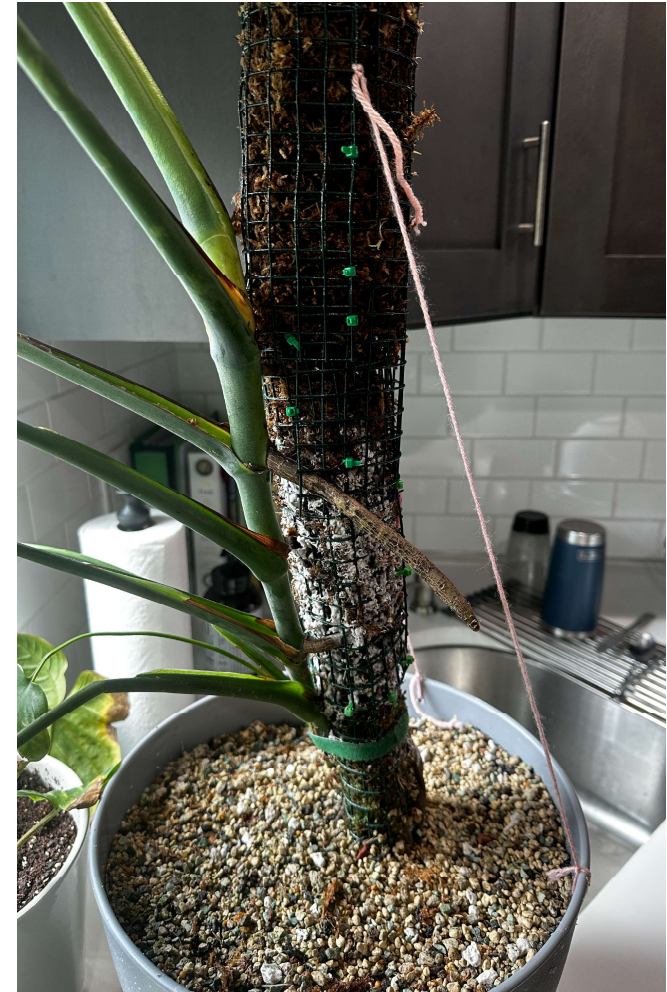
01 Get your shit together



Ideally, you won't need to tie it the whole way up. We'll touch on this later, but the plant should start climbing on its own after the first two ties. If it doesn't, we may have placed it incorrectly.



This Monstera achieved this growth in a single growing season. The moss pole was installed in March and the photo was taken in September; they have a very healthy growing season.



Note how this pole starts thin and was built thicker over time. This is what we're trying to avoid. It's not structurally sound and required anchors to keep stability.

Section 02

Get to know your plant



02 Get to know your plant



Plant Anatomy

Monstera has a front and a back to them, much different than most other plants. The petioles and foliage emerge from the nodes and face forwards while the aerial roots emerge from the back.

Our goal is to place the moss pole at the back of the plant so that the aerial roots will make contact with the moss and grow into it. They'll form a secondary root network in the moss, which is why it's imperative you keep the moss moist.

It can be difficult to spot the stems of younger plants, don't worry if you struggle to. Once your plants are divided, it will become a tad easier.

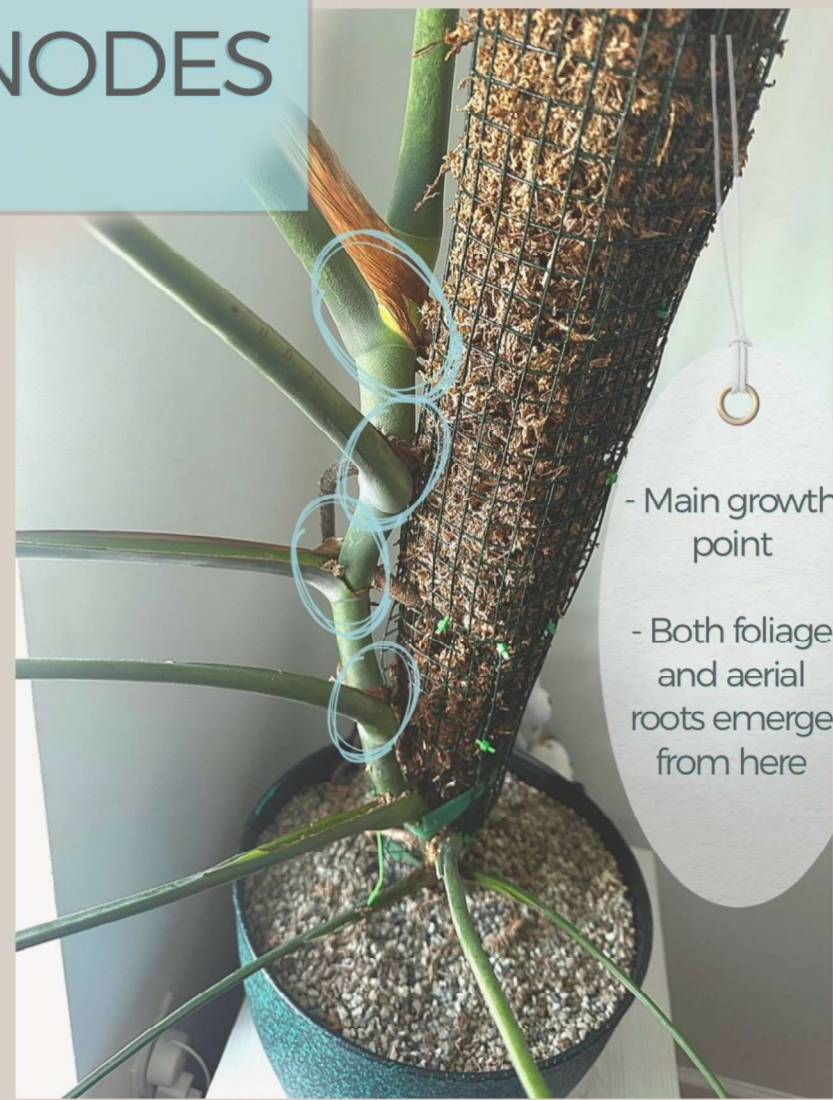
02 Get to know your plant

PETIOLES



- Connects the leaf blade to the stem
- Conduit for water, nutrients, and sugars
- Flexible, allowing the leaves to move and adjust

NODES



- Main growth point
- Both foliage and aerial roots emerge from here

02 Get to know your plant

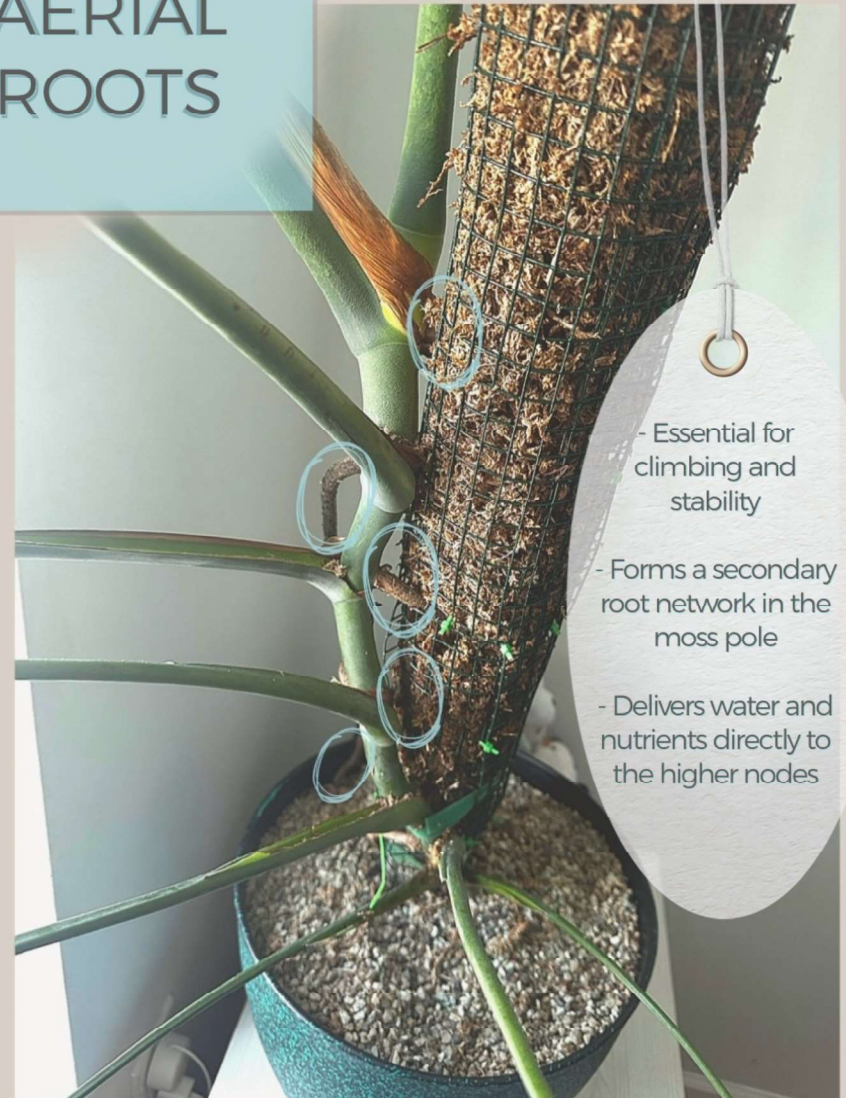
INTER-NODES



- Propagation site of the stem, where you'll make your cuts

- Where you tie the plant if attaching to a moss pole

AERIAL ROOTS



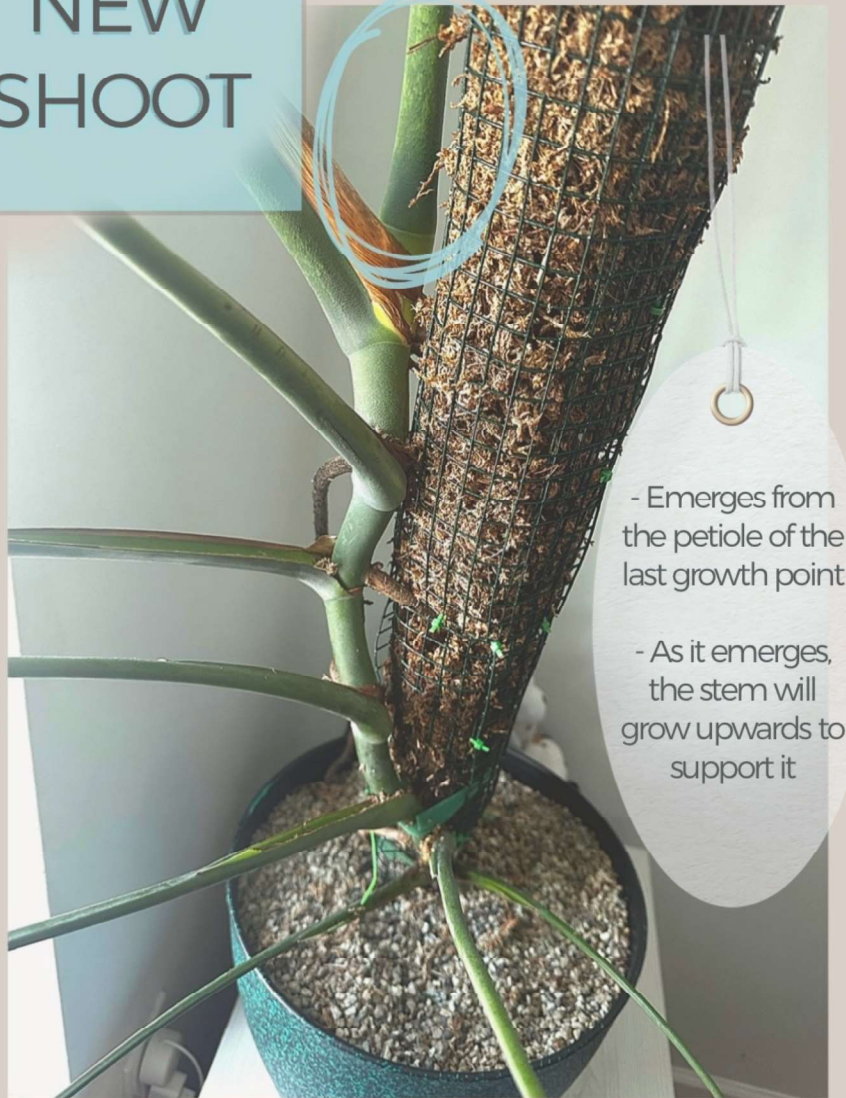
- Essential for climbing and stability

- Forms a secondary root network in the moss pole

- Delivers water and nutrients directly to the higher nodes

02 Get to know your plant

NEW SHOOT



- Emerges from the petiole of the last growth point
- As it emerges, the stem will grow upwards to support it

THE STEM



- The entire structure makes up the stem
- Stores water and nutrients
- Main support

Section 03

Division



03 Division

Dig her up.

Go ahead and pull your plants out of the pot. Don't grab them by the petioles and pull it, just give some gentle tugs to see how difficult it is. If you face resistance, grab your handy chopstick and run it around the perimeter of the pot to loosen up the soil until you can pull it entirely out.

You're likely going to have a very large, very tangled up root ball. Our job now is to make sure each plant retains its own root network; we need to preserve as much of this structure as we can, just untangled.



03 Division

Dry stage.

Our goal in this dry stage is just to let any loose soil fall and remove any obvious amendments. Amendments are your perlite, bark, pumice, anything that's not soil. Depending how much moisture it's retained, it may just fall away on its own. If not, carefully remove them if you can.

Our priority throughout this process is to be gentle. Monstera have very hardy root structures that can take quite a beating, but we're already shocking the plant by doing this and don't need to make things worse.

Move slowly and carefully. This is going to take a lot of time. Be patient throughout, it's worth it.

Some root balls may be coiled up. Use your chopstick to push through any breaks in the roots to try and dislodge a chunk of soil. Usually the center of a coiled root ball is straight soil.



03 Division

Wet Stage.

You probably still have a hefty root ball, that's okay. Grab your plant by the root ball and head to your bathroom. Get the bathtub running at room temperature and a light pressure (if your bathtub is aggressive like mine, the kitchen sink works just fine). Be sure to put a rag in the drain to prevent soil from clogging your pipes while still allowing the water to drain.

Cold water and hot water will both shock your roots, it's important to keep it room temperature and at a light water pressure.

Hold the root ball underneath the water and start working your hands through it, knocking soil away as you do. Get what you can, but it's normal to run into knots in the roots.

If the roots begin to get in your way of clearing the soil out, grab your chopstick. Starting at the top of the root ball, use the chopstick to tug at a root and try to follow it all the way to the bottom. Use the chopstick to help guide it and untangle it as you do.

Very slowly, you'll start to see how each root network is connected to each plant.



03 Division



Wet Stage.

As the roots become more clear, and as you untangle each section more and more, there will come a point when you can pull a plant away.

Pick one of the plants on the outskirts of the root ball and gently tug at it. It'll be a little difficult at first, but once it's dislodged you'll be able to pull it away. Make sure each plant you

Keep moving slowly. The wet stage and division takes us a good hour or two.

If you break some roots along the way, give yourself grace. It's going to happen; we're doing some major surgery to this root ball. But they're hardy and they'll live - it just may take a bit longer to recover is all.

Some plants may only have one or two roots, that's fine. When several plants are in one pot like this, they compete for space and nutrients. This leads to each plant maturing and growing at different rates. You'll usually have one or two plants with large root structures, one or two that are medium, and one or two that are juvenile.

03 Division

The Plug

Once you make your way into the center of the root ball, at the base of the plants you'll almost certainly find a plug. We often refer to these as death plugs, but they were simply used to root the plant as a seedling. We'll need to remove it to split the plants apart, but it'll be a little more difficult.

If it's a mesh plug, grab your pruners and cut sections off as you can. Carefully try to pull the mesh through the roots without ripping them out.

If it's a sponge plug (like to the right), use your chopstick to carefully dislodge small chunks at a time. Too big of a piece and you risk ripping those roots out.



03 Division

Root Rot

To be clear, we never anticipate finding rot. If your plant is healthy, this should be rare. But if you're finding the ends of the roots are black, mushy, and smell like expired eggs, then unfortunately there's an overwatering issue happening and the roots are rotting.

You'll want to cut away any rot. Cut all the way up to the healthy part and leave absolutely zero rot or else it will spread.

Then dunk the root ball into some hydrogen peroxide to disinfect any bacteria that may still be present.

Be aware, your roots will be stained from the soil and look black; don't confuse this with rot. Mushy and a bad smell are your indicators.



Section 04

Planting



04 Planting



Introduce the pole.

With all of your plants separated, it's time to get them into a planter with a pole. Go ahead and put a 1-2" layer of your soil mix into the planter.

Grab a plant and a pole and hold them together in one hand. You want to position the plant against the pole so that aerial roots are able to make contact with it. If you don't have any aerial roots to help you figure it out, look at which direction the petioles are facing and place the pole directly behind them. Don't worry if you're holding onto its petioles, there likely isn't enough stem to hold it to the pole yet. You want to position the plant so that the pole reaches the bottom of the planter, but the plant is sitting at your soil line.

With your free hand, start adding your soil mix to the planter filling in around the plant. You want the part of the pole that's in the planter to be filled with your mix to help anchor it, so add your mix inside the pole as well. This part is difficult if you don't have a friend, but doable.

Once your planter and pole are filled with soil, you won't be able to release the plant yet. We need to tape it up now until it attaches itself to the support.

04 Planting



Fill the pole.

With the plant taped to the pole, you should be able to step away now. Grab a big bowl and fill it with moss. Fill the moss bowl with water and start squeezing it to ensure the moss is completely saturated.

Empty the bowl of water and, one handful at a time, wring out all of the excess water from the moss. It should be soaked, but not dripping. Use this to start filling your pole. Go ahead and fill to the top of your pole. Your plant won't be up there yet, but it likely will be soon so save yourself some time later and just fill it up.

Repeat this process until they're all planted. If you have too many plants, you can put two on one pole. It's not ideal, but they'll certainly grow. Avoid planting more than two in one pot, they're far too large of a plant to adequately have three climbing one pole.

Don't worry if the leaves aren't falling into place right away. Remember, they move around quite a bit. As it goes through shock, it will begin to correct itself and settle into its new home.

04 Planting

Pole Tips

Never use zip ties to attach your plant to the pole. The stem will thicken as it grows and zip ties will cut into it and damage it.

Never restrain your petioles. Monstera's leaves need to be able to move around freely to search for light. If you have the petioles taped up, they'll hinder stem growth and the plant will stop growing. If left like this, it will start to decay.

It can be especially difficult to support juvenile plants. The stem will not be visible until the plant has more than one leaf. If you only have the one leaf, aim your tape just below where it emerges like our Monstera Adansonii on the right.

When you tape it up, your leaves will no longer stand straight up like before. This is normal, this is the goal. They'll actually be more horizontal now, which can look a little unruly with only one or two leaves. Fight the urge to stake them or tie them to the pole; we don't want them to be upright anymore. Instead, we need that area clear so that the stem can grow upwards and grow more foliage as it does.



04 Planting



What now?

Now that it's all over, get your plants in their permanent spots. It's important not to mess with them at all anymore while they go through shock. A normal shock period is anywhere from 2-4 weeks; but if you're doing this in Fall or Winter, it could be longer.

As your plant grows, the aerial roots will form an entirely separate secondary root network inside of the moss. This pole will become a permanent fixture for your plant as it attaches and will be carried with it through every repot.

The additional root network will improve water uptake and nutrient distribution leading to even larger leaves and eventually those iconic fenestrations.

If aerial roots miss the pole (it happens), just guide them down into the soil. They won't help the plant climb, but they'll root into the soil too and grab water/nutrients from there.

If all of your aerial roots are missing the pole, it's like the plant is angled wrong. This will require a repot to fix so that you can reorient the plant, but let it grow in before you do this so that you can more accurately see how the stem is forming.

Section 05

Maintenance



05 Maintenance



What now?

You'll need to keep the moss moist while avoiding overwatering the soil below. Remember, any excess water from the moss will drain into your planter. So if you've recently watered the plant, you'll want to avoid watering the pole or risk rotting the roots. Ideally, you'll water both the pole and the plant at the same time.

In between waterings, you can use a mister to mist the moss. This will moisten it without providing too much moisture.



Misters can be expensive. I took a 20oz. Root Beer bottle and poked a few holes in the cap instead. Now, I fill the bottle with water, put the cap back on, and flip it upside down and into the pole. This creates a slow feed of water to the moss which allows you to watch how moist your moss is getting. Once the water reaches the bottom of the pole, remove the bottle before more water is released.